Practice following problems for the preparation of Induction Motor and Synchronous Motors

End Problems: (Chapman textbook 5th edition)

Induction Motor: 6.1 to 6.5

Synchronous Motor 5.1 to 5.3

Also try following problems:

A 440-V 50-Hz two-pole Y-connected induction motor is rated at 75 kW. The equivalent circuit parameters are;

$R_1 = 0.075\Omega$, $R_2 = 0.065\ \Omega$, $X_M = 7.2\ \Omega$, $X_1 = X_2 = 0.17\ \Omega$,

$P_{Mech} = 1$ KW, $P_{Misc} = 150$ W, $P_{Core} = 1.1$ KW

For a slip of 0.04, find

1. Input line current
2. Output power
3. The power factor
4. The stator copper losses P_{SCL}
5. The air-gap power P_{AG}
6. The induced torque τ_{ind}
7. Efficiency

A 208-V, Y-connected synchronous motor is drawing 40 A at unity power factor from a 208-V power system. The field current flowing under these conditions is 2.7 A. Its synchronous reactance is 0.8 Ω. Assume a linear open-circuit characteristic(i.e. field current is directly proportional to the generated voltage).
(a) Find the torque angle δ.
(b) How much field current would be required to make the motor operate at 0.8 PF leading (for the same output power)?
(c) What is the reactive power of motor ***in part (b)****?*

A 400V, 60 HP, 60 Hz, 8 pole, Y connected Synchronous motor has per phase synchronous reactance of 0.4 Ω. The motor is initially working at 0.8 pf lagging and delivering an output power of 32000 W. The field current of the motor is changed so that the motor is now operating at a pf of 0.7 leading keeping output power constant. The field circuit of the motor has a voltage source of 250 V (DC), and all the losses of the motor are ignored. The motor is operating at rated voltage and frequency.

Find out following

(a) Field current in the initial case at 0.8 lagging pf

(b) The value of total field resistance in final case at 0.7 leading pf

(c) Induced torque of the motor in the final case

The OCC is shown below

I_F (A)	2	4	6	8	10
$V_{T,NL}$ (V)	200	400	500	550	600